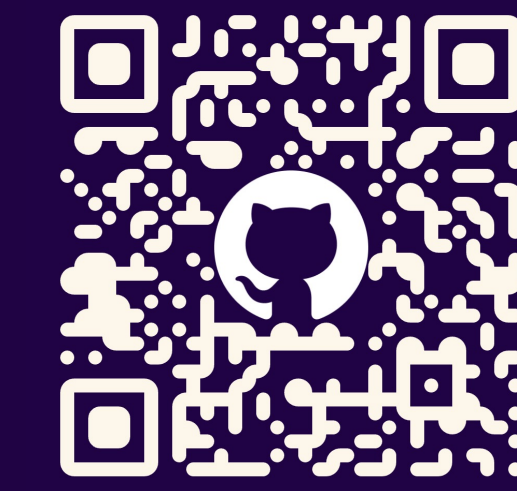




Introduction

Future hydrogen-powered jet engines will require compact and low pressure drop heat exchangers to enable new and advanced cycles. The transition to liquid hydrogen as a commercial aviation fuel offers a promising pathway to low-carbon air transport and the opportunity for substantial reductions in take-off weight of long-haul aircraft. However, these specific aerospace applications necessitate tube bank designs with uniquely large transverse spacings, up to 6 diameters between the tube rows, which fall far outside the geometric boundaries of existing empirical correlations.

To bridge this knowledge gap, this research project investigates the flow behaviour and performance of tube banks at extended transverse spacings using 2D Unsteady Reynolds-averaged Navier-Stokes (URANS) computational fluid dynamics (CFD) simulations.

Background

- **Hydrogen Aviation:** Liquid hydrogen produces three times more energy per kilogram than kerosene while producing no CO₂ emissions at the point of use.
- **Extended Spacing Challenge:** Low pressure drop requirements in the Lockheed hydrogen aircraft studies of the 1970s and 1980s used wide tube spacings ($X_t^* = 6$), outside the range of validity of existing correlations.
- **Literature Limitation:** Standard correlations (Kays & London $X_t^* = 1.5$, Gaddis & Gnielinski $X_t^* \leq 3$) overpredict performance when extrapolated.
- **Numerical Strategy:** 2D URANS balances computational cost and resolves transient vortex shedding across multi-lane domains.

Mesh Independence Study

A mesh independence study was performed to calculate the minimum mesh size required, in order to reduce the run time of simulations, while still providing results that minimise the errors introduced from the mesh. The base cell size was decreased from $d_o/2$ to $d_o/10$, with the flow parameters were evaluated and compared to the finest mesh produced, Fig.1. It was desirable for the mesh to have a percentage difference of less than 1% when compared to the finest mesh, as such the base size of $d_o/5$ was chosen, shown in Fig.3. This mesh size increased the run time of the simulations for the $d_o/5$ mesh to 6 hours compared to 3 hours for the $d_o/2$ mesh.

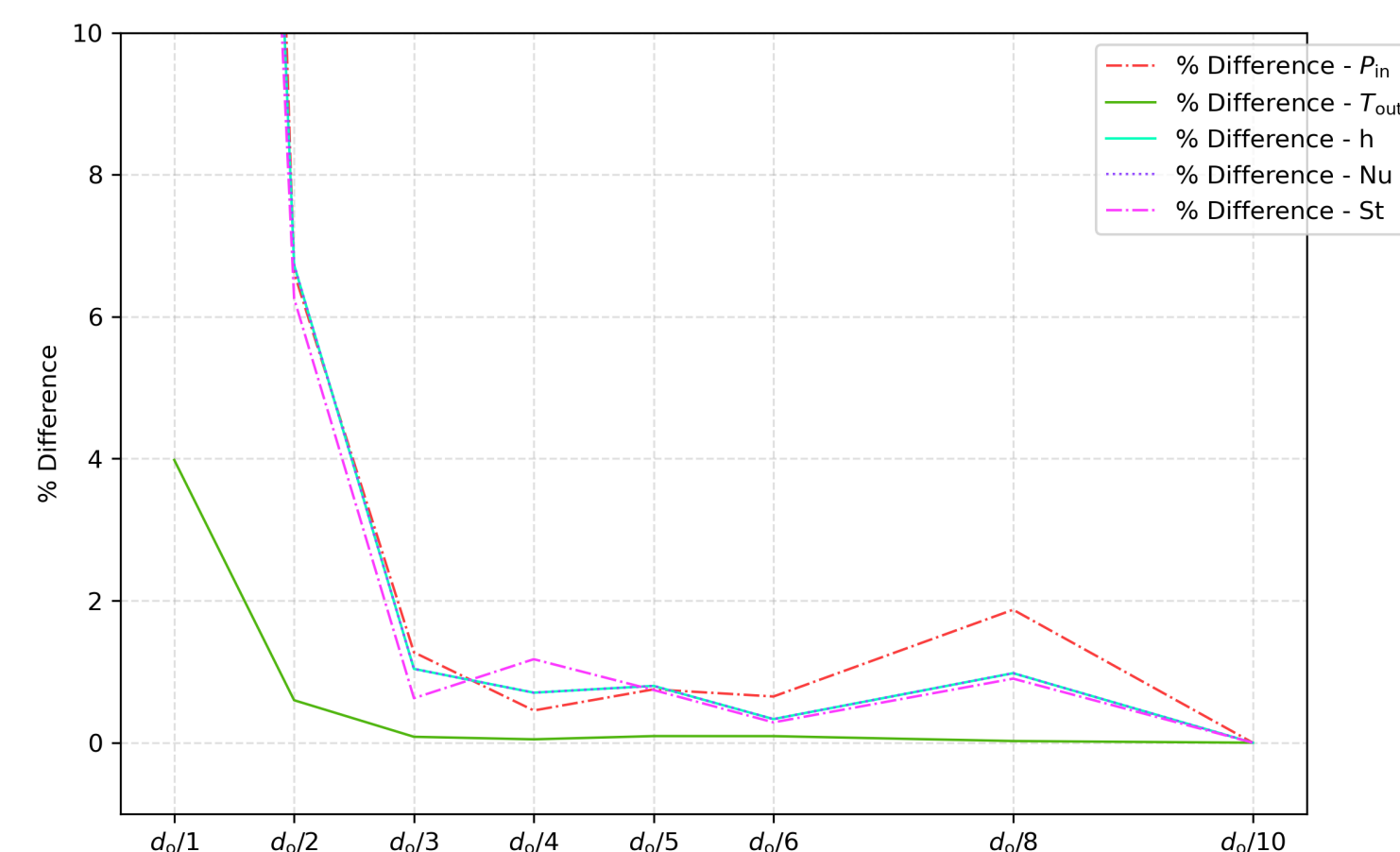


Figure 1. Percentage difference of flow parameters compared to finest mesh.

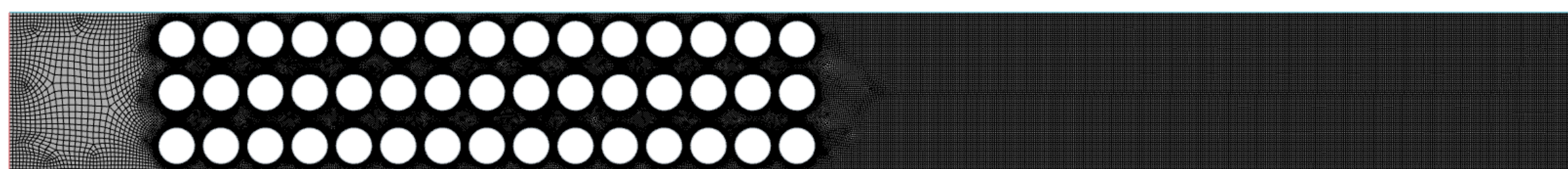


Figure 3. $d_o/5$ base cell size mesh.

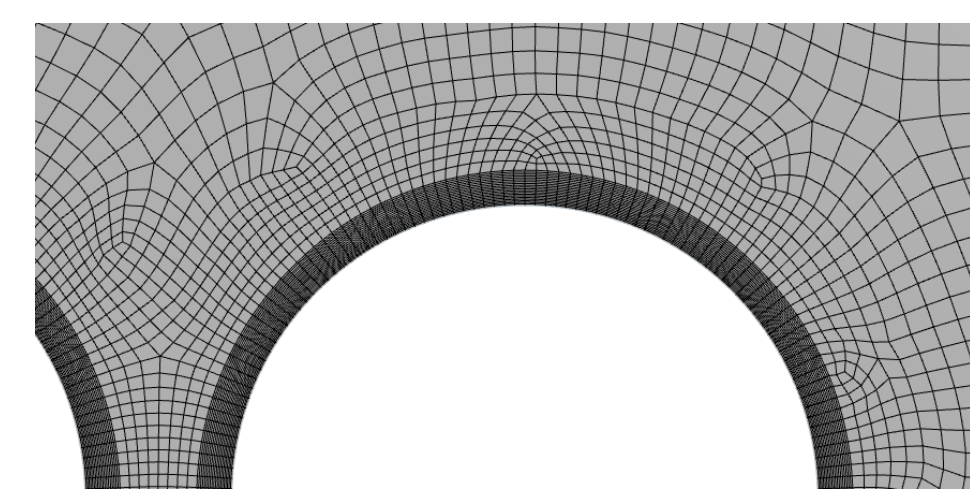


Figure 2. Prism layers of $d_o/5$ mesh around the tube walls.

Simulation Setup

To model the tube bank in CFD, the boundary conditions of the domain were set to replicate physical experiments that were done by Kays and London, however, using a different geometry of heat exchanger, shown in Fig.4, with the geometric and boundary condition parameters of the domain also detailed in Table.1.

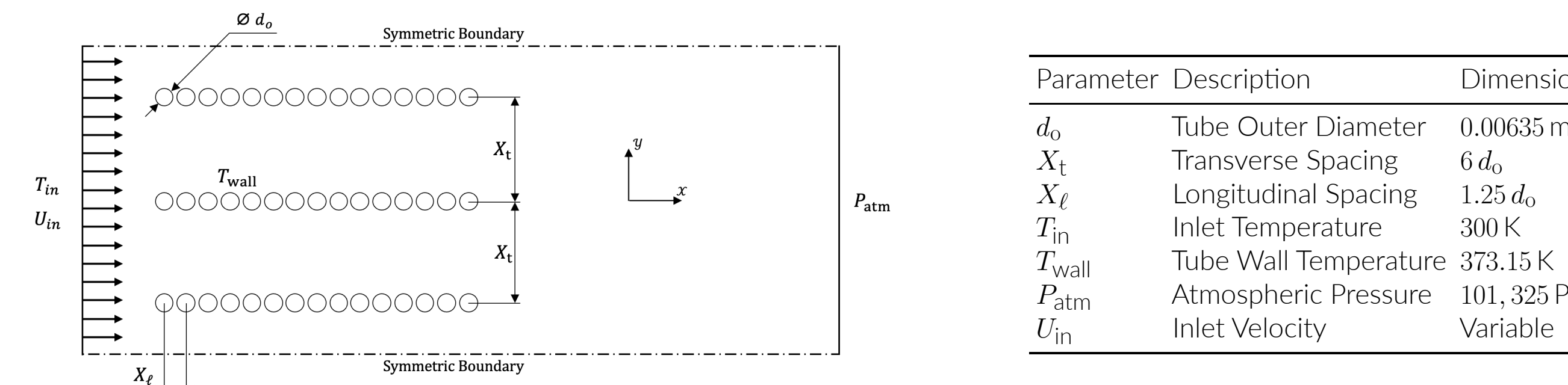


Table 1. Domain parameters.

Figure 4. Diagram of the simulation domain.

Results and Discussion

The simulation results for the extended transverse spacing of $X_t = 6 d_o$ are plotted below in Fig.5 showing the dimensionless heat transfer and Fig.6 showing the dimensionless viscous shear.

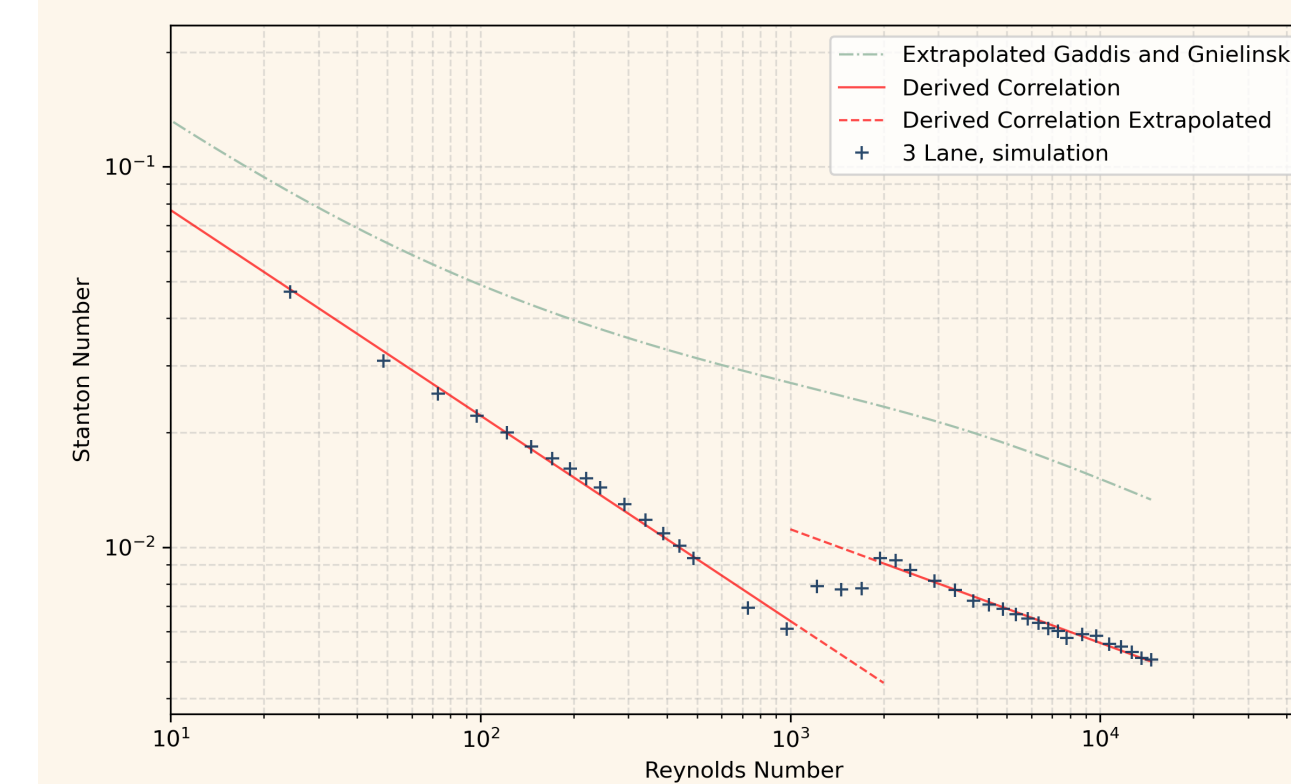


Figure 5. Dimensionless heat transfer

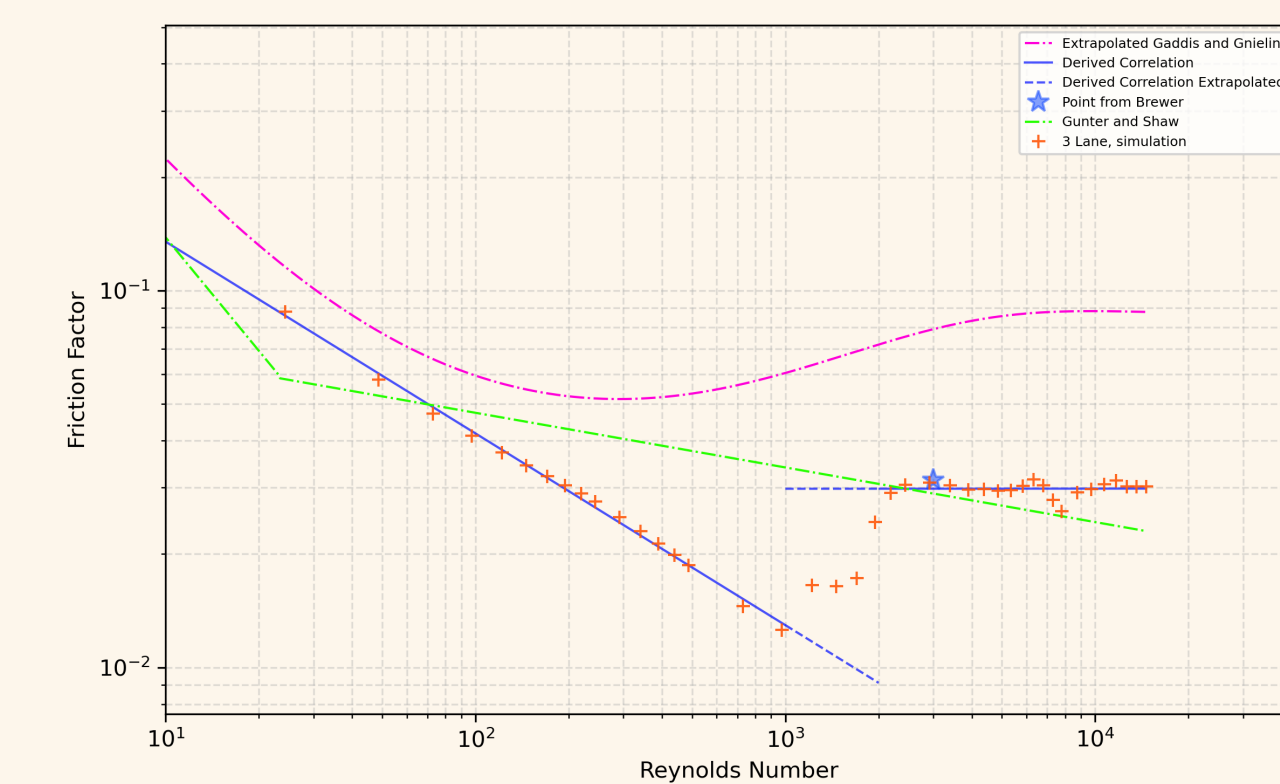


Figure 6. Dimensionless viscous shear (friction)

It is clear that the extrapolated Gaddis and Gnielinski correlation is not good at predicting the heat transfer of friction for the wide spacing, which was to be expected as these should only be used for transverse spacings up to $X_t^* = 3$. However, the generic Gunter and Shaw correlation for the friction is incidentally quite accurate at the Reynolds number of the Brewer design point.

To compliment the literature, it was desirable to produce a set of correlations that could be used for calculating the Stanton number and friction factor across different Reynolds numbers. After computing the derived correlations, they were plotted alongside the simulation data, with the Stanton number correlation plotted in Fig.5, and the friction factor correlation plotted in Fig.6.

$$St(Re_d, Pr) = \begin{cases} 0.298 Re_d^{-0.540} Pr^{1/3}, & 20 \leq Re_d \leq 1,000 \\ 0.0980 Re_d^{-0.299} Pr^{1/3}, & 2,000 \leq Re_d \leq 10,000 \end{cases} \quad (1)$$

$$f(Re_d) = \begin{cases} 0.435 Re_d^{-0.509}, & 20 \leq Re_d \leq 1,000 \\ 0.0297, & 2,000 \leq Re_d \leq 10,000 \end{cases} \quad (2)$$

To better understand the unsteady flow behaviour for incline tube banks, a 3 lane simulation was ran at different Reynolds numbers, these are shown in Fig.7. In the progression of Reynolds number, you can see the change from structured vortex streets visible at $Re_d = 150$ to the transitional at $Re_d = 1,460$, to the final high Reynolds case at $Re_d = 14,600$ where the flow around the tube bank has oscillatory structure forming before the wake and within the tube bank. This analysis was useful to understand the difficulties of using CFD for tube bank heat exchanger predictions.

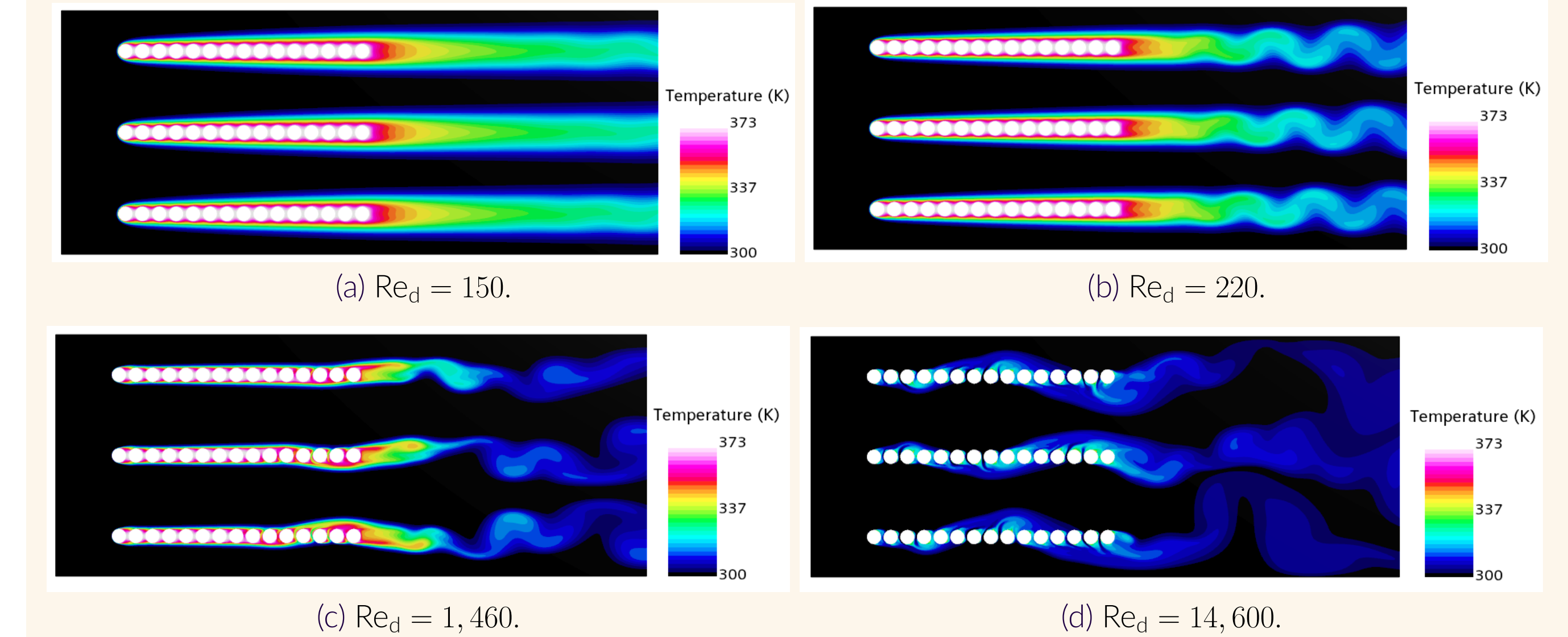


Figure 7. Inline $X_t^* = 6$, $X_l^* = 1.25$ temperature scenes for different Reynolds number.

Conclusion

This research project evaluated heat exchanger performance at extended transverse spacings ($X_t^* = 6$) for recuperated hydrogen turbofan aircraft:

- Turbulent URANS modelling with at least three tube lanes was shown to capture the increase in friction factor and Stanton number seen in low transverse spacing published experiments: vortex streets and coalescing jets are the underlying physical structure and instability.
- The three-lane simulations successfully replicated the single data point from Brewer's hydrogen aircraft research, which was also well predicted by Gunter and Shaw's correlation for both transverse and inline tube banks.
- That correlation did not however account for inline tube bank's characteristic transition region of low friction factor and Stanton number which is shown in experimental data by Kays and London. The region was found to have a lower friction factor by about 0.01 to 0.02, corresponding to a 25% to 50% lower value than that predicted by the correlation.
- New piecewise log-linear correlations, Eqn.1 and Eqn.2, were successfully derived from the simulation data. These correlations offer predictive models tailored specifically for inline configurations with a transverse spacing of $X_t^* = 6$.

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